

120 Days of Quantitative Finance

Month 4: Stochastic Calculus & Time Series Modelling

Week 15: Advanced Time Series Modelling

Day 105: Historical vs Implied Volatility

1 Introduction

Volatility is a key concept in finance, representing the uncertainty or risk of an asset's returns.

There are two primary types:

- Historical (realized) volatility
- Implied volatility

2 Historical Volatility

Historical volatility is computed from past returns.

$$\sigma_{hist} = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (r_t - \bar{r})^2}$$

2.1 Interpretation

- Measures past market variability
- Backward-looking
- Based on realized data

3 Implied Volatility

Implied volatility is derived from option prices using models like Black-Scholes.

It represents the market's expectation of future volatility.

3.1 Interpretation

- Forward-looking
- Reflects market sentiment
- Extracted from option prices

4 Key Differences

- Historical volatility uses past data
- Implied volatility reflects future expectations
- Implied volatility often exceeds historical volatility

5 Volatility Risk Premium

Typically:

$$\text{Implied Volatility} > \text{Historical Volatility}$$

This difference is known as the volatility risk premium.

6 Financial Interpretation

- Implied volatility includes risk aversion
- Historical volatility reflects realized uncertainty

7 Applications

- option pricing
- volatility trading strategies
- risk management

8 Conclusion

Understanding the relationship between historical and implied volatility is essential for interpreting market expectations and identifying trading opportunities.